

Lecture 6 (9/25)

Variance, Normals models and Linear Regression

Variance and standard deviation

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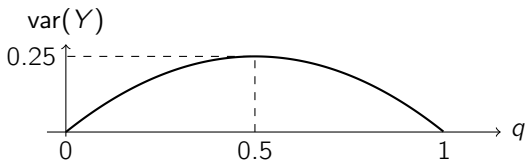
In code `np.var(y)`.

Example

Let $Y \sim \text{Bernoulli}(q)$. Compute $\text{var}(Y)$.

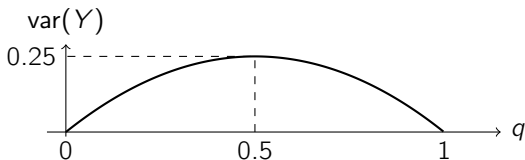
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Let $Y \sim \text{Uniform}(0, L)$. Compute $\text{var}(Y)$.

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$$\sigma = \sqrt{\text{var}(Y)} = \sqrt{E[(Y - E[Y])^2]}.$$

Unlike the variance, σ has the same units as Y , so it tells us the *typical size* of the residual $Y - E[Y]$.

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- The coefficient of variation, or CV , compares the standard deviation to the mean:

$$CV = \frac{\sigma}{\mu}$$

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Note: The CV makes sense for positive random variables. For example, it is meaningless for something like temperature.

The Normal distribution

$Y \sim \text{Normal}(\mu, \sigma^2)$ has density

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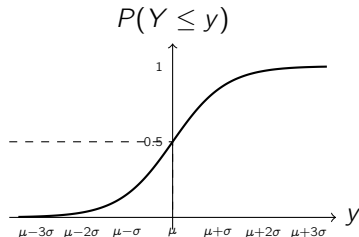
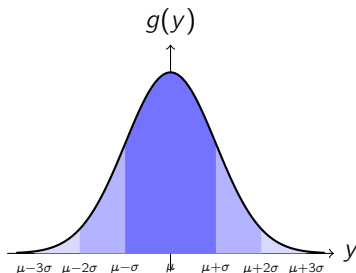
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- If X_1, X_2 are independent Normal and a, b, d are constants, then

$$aX_1 + bX_2 + d \sim \text{Normal}(a\mu_1 + b\mu_2 + d, a^2\sigma_1^2 + b^2\sigma_2^2)$$

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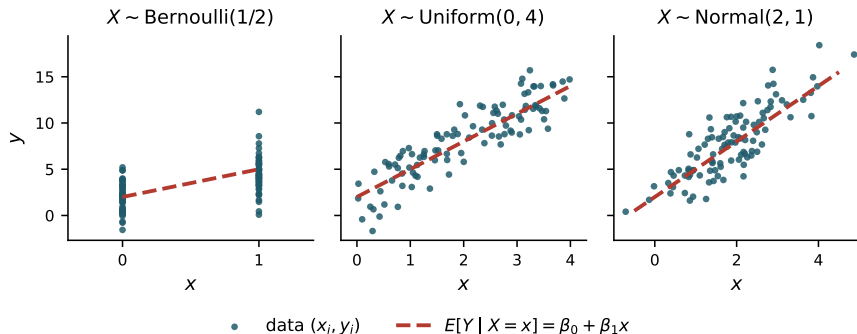
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- X is the predictor
- Y is the response variable
- β_1 is called the regression coefficient of X .

What data from this model looks like

Data simulated from $Y | X \sim \text{Normal}(\beta_0 + \beta_1 X, \sigma^2)$



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- Exogeneity: ϵ is independent of X .

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$$\text{cov}(X, Y) = E[XY] - E[Y]E[X]$$

and from the calculations above

$$\beta_1 = \frac{\text{cov}(X, Y)}{\sigma_X^2}$$

Correlation coefficient